

Embedded Linux & System Software Interview Handbook

Chapter 6 - Device Tree & Platform Drivers (Part 3)

Topics: Writing Complete Device Tree Nodes for Common Peripherals

6.6 Complete Device Tree Examples

The Device Tree describes every hardware peripheral available on a board. During boot, Linux parses these nodes and passes the configuration to the appropriate platform driver.

Peripheral	Important Properties
UART	compatible, reg, interrupts, clocks, status
GPIO	gpio-controller, #gpio-cells
I2C	clock-frequency, #address-cells
SPI	num-cs, cs-gpios
Ethernet	phy-handle, local-mac-address
PWM	pwms property
Watchdog	timeout-sec
Fixed Regulator	regulator-name, regulator-min-microvolt

Example UART Node

```
uart0: uart@1000 {
compatible = "vendor,uart";
reg = <0x1000 0x100>;
interrupts = <5>;
clocks = <&uart_clk>;
status = "okay";
};
```

Example I²C Bus

```
i2c0 {
clock-frequency = <400000>;
temperature@48 {
compatible="ti,tmp102";
reg=<0x48>;
};
};
```

Example SPI Device

```
spi0 {
flash@0 {
compatible="jedec,spi-nor";
reg=<0>;
spi-max-frequency=<50000000>;
};
};
```

```
};
```

Pin Control (pinctrl)

The pinctrl subsystem configures the electrical function of SoC pins. Drivers usually reference pinctrl-0 and pinctrl-names to apply the required pin multiplexing before accessing hardware.

How Linux Uses the Device Tree

1. Bootloader loads Kernel + DTB.
2. Kernel parses the DTB.
3. Platform devices are created.
4. Compatible strings are matched.
5. Driver probe() configures the hardware.

Board Bring-up Checklist

- Verify DTB loaded by U-Boot.
- Check /proc/device-tree.
- Inspect dmesg.
- Validate compatible strings.
- Confirm clocks, GPIOs and interrupts are correctly defined.

Interview Questions

1. How is an I²C device represented in the Device Tree?
2. Why is pinctrl required?
3. What happens if a compatible string is incorrect?
4. How are child devices represented on an I²C or SPI bus?
5. Which files help verify the running Device Tree?

Key Takeaways

A well-written Device Tree enables reusable drivers across multiple boards. Correct peripheral descriptions are critical for successful board bring-up and Linux platform driver initialization.